Introduction to computers

Computer

- An electronic device that processes data according to a set of stored instructions.
- An electronic device that follow instructions to accept input, process the input and then produce information.
- A machine that manipulates data according to a set of instructions.
- A machine that perform tasks or calculations according to a set of instructions or programs.

Computers

- Work through an interaction of hardware and software
  - **Hardware** refers to physical parts of a computer that you can see and touch, including the case and everything inside it
  - **Software** refers to programs that governs the operations of the computer.

Computer hardware

- These are physical parts of a computer that one can see and touch, including the case and everything inside it
- Hardware items such as monitor, keyboard, mouse, printer and other components are called hardware devices.
  - Internal hardware
    - Located inside the main box (system unit) of the computer
  - External hardware
    - Located outside the system unit and plug into ports located on the exterior of the system unit

Software

- In addition to hardware, a computer needs software to function.
- Are programs that control or maintain the operations of the computer and its devices.
- *Software* or *programs,* consists of electronic instructions that tell the computer how to perform a task.
- A program installed in the computer that contains a set of instructions executed by the computer in order to perform a task.
- Consist of computer programs that govern the operation of the computer.
- Software is needed to control hardware and to turn it into a useful machine.

Types of software

- **System software**:
  - Programs that control or maintain the operations of the computer and its devices.
  - Programs that controls a computer system.
    - **E.g. 1. Operating System** - controls and manages the hardware connected to a computer
    - OS - provides an interface that helps users to interact with the computer.
    - Boots the computer and launches programs at the user's direction
      - Examples of OS are Windows XP, 7,8,10, Mac OS, Linux, Android etc.
    - **E.g. 2. Utility programs** – A program that performs a very specific task related to managing system resources.
    - Help to perform maintenance or correct problems with a computer system.
      - E.g. Antivirus – utilities scan for computer virus, backup software, data compression, system monitors etc.

Types of software

2. **Application software**: Programs that perform specific tasks for users.
   - *Application software* enables users to perform specific tasks—solve problems, perform work, or entertain yourself.
     - E.g. - Word Processing for creating letters, reports, etc.
     - Spreadsheets – for creating budgets etc
     - Scheduling appointments
     - Viewing Web pages
     - Sending and receiving e-mail

Types of computers

Desktop computers

- Designed for use at a desk or

Laptops and notebooks

- Laptops are lightweight mobile PCs with a thin screen.
- Laptops can operate on batteries, so you can take them anywhere.
- Unlike desktops, laptops combine the CPU, screen, and keyboard in a single case
- Notebooks are small, affordable laptops that are designed to perform a limited number of tasks such as to browse the web and check e- mail

Handheld computers (Palmtops)

- Also called personal digital assistants (PDAs)
- Are battery-powered computers small enough to carry almost anywhere.
- Are useful for scheduling appointments, storing addresses and phone numbers, and playing games.
- Some have more advanced capabilities, such as making telephone calls or accessing the Internet.
- Instead of keyboards, handheld computers have touch screens that you use with your finger or a stylus (a pen-shaped pointing tool)

Tablet PCs

- Are mobile PCs that combine features of laptops and handheld computers.
- Like handheld computers, users write notes or draw pictures on the screen with a tablet pen instead of a stylus.
- Can also convert handwriting into typed text.
- Some Tablet PCs are “convertibles” with a screen that swivels and unfolds to reveal a keyboard unde

Smartphones

- Are mobile phones that have some of the same capabilities as a computer.
- Used to make telephone calls, access the Internet, organize contact information, send e- mail and text messages, play games and take pictures.
- Usually have a keyboard and a large screen.

Server

- A computer that has been optimized to provide services to other computers over a network.
- Servers have powerful processors, lots of memory and large hard drives.
- Other types of computers can fill an entire room.
- These are: ***Mainframes and Supercomputers***

What can you do with computers?

- At workplace, people use computers to keep records, analyze data, do research, and manage projects.
- At home, people use computers to find information, store pictures and music, track finances, play games and communicate with others.
- Connect to the Internet etc.

Components of a computer

- Central Processing Unit (CPU)
- Main memory
- Storage devices
- Input devices
- Output devices

Central Processing Unit (CPU)

- Also called the processor or microprocessor.
- A part of a computer that controls all other parts.
- CPU interprets and carries out the basic instructions that operate a computer.
- Directs and coordinates most of the operations in a computer
- Works hand in hand with other circuits such as *main memory* to carry out processing.
- Considered to be the brain of the computer.

Memory

- A part of a computer in which information or programs are stored either permanently or temporarily.
- Main memory is where programs and data are kept when the processor is actively using them.
- Memory is a temporary storage of data, instructions and information
- Special chips that hold data and instructions ready for use by the CPU.

Types of memory

- Random Access Memory (RAM)
  - Also referred to as Main memory
  - Holds the programs and data that users are working on. (Are stored in RAM)
  - **Temporarily hold (1) software instructions and (2) data before and after it is processed by the CPU.**
  - RAM is volatile – when the computer is switched off anything in RAM is lost.
  - Data stored on disk is transferred into RAM when you open a file to work on it.

- **Read Only Memory (ROM)**
  - Memory that stores instructions and data that can be only be read, but it cannot be updated or modified.
  - Modifying it is either impossible or very difficult.
  - Important program instructions needed to start a computer and load OS are stored in ROM.
  - ROM is **non-volatile** storage, which means that the information is maintained even if the component loses power.
  - ROM plays a critical part in starting up the computer.
    - This is also referred to as **booting up** the computer.

Hard disk drive (HDD)

- HDD is storage device of the computer
- Contains computer programs and data
- Most computer are sold with a hard disk drive built into them.
- Additional HDD can be bought to increase storage space (External or Internal)
- External HDD more expensive than internal
- Cost of HDD increases with its capacity
- HDDs store between 80GB and 500GB

Input devices

- Any device that enables users to enter data and instructions into your computer
  - Keyboard
    - Used to type information into your computer
  - Mouse
    - Allows data input by clicking to select options (rather than typing).
  - Scanners
  - Barcode readers
  - Digital camera
  - Microphone

Output Devices

- Any device that help users get information out of the computer in a meaningful way.
  - Monitor, screen or Visual Display Unit (VDU)
    - Cathode Ray Tube (CRT)
      - A vacuum tube used as a display screen in a computer or video display terminal.
      - Found in older desktop computers but also in older television sets.
    - Liquid Crystal Display (LCD)
      - A flat panel technology
  - Printers
  - Speakers
    - To deliver an audio output such as music, that is connected to the computer.